

Data Visualization

60 questions · compiled from 1 file

DATA VISUALIZATION IN DATA ANALYSIS

1. Which statement best explains why visualization is valuable alongside summary statistics?

1. Visuals can reveal patterns that summary numbers may hide
2. Summary statistics are never useful
3. Visuals always provide exact probabilities
4. Visuals eliminate sampling bias

Answer 1 · Visuals can reveal patterns that summary numbers may hide

DATA VISUALIZATION IN DATA ANALYSIS

2. What is the best reason to iterate through multiple visualizations during exploration?

1. To discover different structures and relationships in the data
2. To make the final report longer
3. To avoid cleaning the dataset
4. To guarantee a significant result

Answer 1 · To discover different structures and relationships in the data

DATA VISUALIZATION IN DATA ANALYSIS

3. Which of the following is most likely an exploratory task?

1. Testing several plot types to understand a variable
2. Creating a polished infographic for executives
3. Finalizing a presentation theme
4. Writing a press release about findings

Answer 1 · Testing several plot types to understand a variable

DATA VISUALIZATION IN DATA ANALYSIS

4. Why should analysts avoid locking into one narrative too early during exploration?

1. Early assumptions can bias what patterns they notice and communicate
2. It makes Python code run slower
3. It prevents charts from rendering
4. It changes the raw data values

Answer 1 · Early assumptions can bias what patterns they notice and communicate

DATA VISUALIZATION IN DATA ANALYSIS

5. What is a major benefit of visualization when checking data quality?

1. It can help identify outliers, missingness, and unusual patterns
2. It automatically fixes all data errors
3. It proves the dataset is unbiased
4. It replaces domain expertise

Answer 1 · It can help identify outliers, missingness, and unusual patterns

6. Which situation best calls for an explanatory rather than exploratory chart?

1. Presenting a final recommendation to stakeholders
2. Trying different axis scales to inspect shape
3. Checking whether a variable is skewed
4. Looking for possible clusters in raw data

Answer 1 · Presenting a final recommendation to stakeholders

7. What is the role of visualization after analysis is complete?

1. To communicate findings effectively
2. To recollect missing data
3. To define the original business problem
4. To replace written conclusions entirely

Answer 1 · To communicate findings effectively

8. Which statement is most accurate about visualization in decision-making?

1. Good visualizations support decisions by making evidence easier to interpret
2. Visualizations guarantee correct decisions
3. Only dashboards can support decisions
4. Visualizations are useful only for technical audiences

Answer 1 · Good visualizations support decisions by making evidence easier to interpret

9. Why might an analyst pair tables with charts in a report?

1. Charts show patterns while tables provide precise values
2. Tables are always more persuasive than charts
3. Charts cannot display categories
4. Tables remove the need for interpretation

Answer 1 · Charts show patterns while tables provide precise values

10. What is the strongest reason to keep exploratory charts even if they are not presentation-ready?

1. They help analysts reason through the data and refine questions
2. They are always suitable for executives
3. They reduce the need for documentation
4. They guarantee reproducibility without code

Answer 1 · They help analysts reason through the data and refine questions

11. Which design choice most improves readability in a chart?

1. Clear labels and a simple layout
2. Maximum color saturation everywhere
3. 3D bars with shadows
4. Decorative backgrounds

Answer 1 · Clear labels and a simple layout

12. Why is position on a common scale considered a strong visual encoding?

1. People compare aligned positions more accurately than many other encodings
2. It makes legends unnecessary
3. It works only for categorical data
4. It always reduces chart size

Answer 1 · People compare aligned positions more accurately than many other encodings

13. Which chart design is most likely misleading?

1. A chart with a distorted axis scale
2. A chart with labeled units
3. A chart with a descriptive title
4. A chart with a legend for categories

Answer 1 · A chart with a distorted axis scale

14. What is a key reason to use consistent scales across comparable charts?

1. It enables fair visual comparison between panels or groups
2. It makes all variables normally distributed
3. It removes the need for annotations
4. It guarantees the same sample size

Answer 1 · It enables fair visual comparison between panels or groups

15. Why should axis labels include units when relevant?

1. To make the values interpretable
2. To reduce the number of ticks
3. To increase decoration
4. To avoid using titles

Answer 1 · To make the values interpretable

16. Which practice best supports accessibility in visualization design?

1. Using distinguishable colors and non-color cues where possible
2. Encoding all meaning with color alone
3. Using tiny text to fit more detail
4. Removing labels to reduce clutter

Answer 1 · Using distinguishable colors and non-color cues where possible

17. When is a legend preferable to direct labels?

1. When direct labeling would create clutter or is impractical
2. Always, regardless of chart type
3. Never under any circumstances
4. Only when there is one category

Answer 1 · When direct labeling would create clutter or is impractical

18. Why can pie charts become ineffective with many categories?

1. Small slices are hard to compare accurately
2. They cannot show proportions
3. They always require logarithmic scales
4. They are only for time series

Answer 1 · Small slices are hard to compare accurately

19. What is the main purpose of using a meaningful sort order in bar charts?

1. To make comparisons easier
2. To increase chart complexity
3. To hide low values
4. To avoid labeling categories

Answer 1 · To make comparisons easier

20. Why should designers be cautious with dual-axis charts?

1. They can imply misleading relationships if scales are not interpreted carefully
2. They cannot display numeric data
3. They are required for all comparisons
4. They automatically normalize variables

Answer 1 · They can imply misleading relationships if scales are not interpreted carefully

21. Which plot is especially useful for seeing the shape of a numeric distribution?

1. Histogram
2. Scatter plot
3. Sankey diagram
4. Choropleth map

Answer 1 · Histogram

22. What does a bimodal histogram suggest?

1. The data may contain two clusters or subpopulations
2. The variable is definitely categorical
3. There are no outliers
4. The mean equals the median

Answer 1 · The data may contain two clusters or subpopulations

23. What does the spread of a distribution describe?

1. How dispersed the values are
2. How many categories exist
3. Whether the chart has labels
4. The file size of the dataset

Answer 1 · How dispersed the values are

24. Why might a box plot hide some distribution details compared with a histogram?

1. It summarizes data into quartiles and whiskers rather than showing full shape
2. It cannot show outliers
3. It changes the underlying values
4. It only works for categorical variables

Answer 1 · It summarizes data into quartiles and whiskers rather than showing full shape

25. Which measure is most affected by extreme outliers?

1. Mean
2. Median
3. Mode
4. Interquartile range only

Answer 1 · Mean

26. If a variable has many repeated values at zero and a long positive tail, which description fits best?

1. Zero-inflated and right-skewed
2. Symmetric and uniform
3. Bimodal and left-skewed
4. Ordinal and balanced

Answer 1 · Zero-inflated and right-skewed

27. Why might you inspect value counts for a categorical variable?

1. To understand category frequency
2. To estimate linear correlation
3. To compute regression coefficients
4. To remove all rare categories automatically

Answer 1 · To understand category frequency

28. What is a likely consequence of using too few bins in a histogram?

1. Important structure may be oversmoothed and hidden
2. The chart becomes a scatter plot
3. The data becomes normally distributed
4. Outliers are automatically removed

Answer 1 · Important structure may be oversmoothed and hidden

29. Which plot is often useful for comparing category frequencies directly?

1. Bar chart
2. Line chart with continuous x-axis only
3. Scatter plot
4. Contour plot

Answer 1 · Bar chart

30. Why is skewness important in univariate analysis?

1. It affects interpretation of center, spread, and suitable transformations
2. It determines whether labels are needed
3. It guarantees causation
4. It only matters for categorical variables

Answer 1 · It affects interpretation of center, spread, and suitable transformations

31. What does a scatter plot primarily help reveal?

1. Association between two numeric variables
2. Distribution of one categorical variable
3. Exact probability values
4. SQL join structure

Answer 1 · Association between two numeric variables

32. If a scatter plot shows a curved pattern, what does that imply?

1. The relationship may be nonlinear
2. There is no relationship
3. The variables are categorical
4. The data must be erroneous

Answer 1 · The relationship may be nonlinear

33. Which chart is suitable for comparing average numeric values across categories?

1. Bar chart with summary statistic
2. Histogram
3. Heatmap of missing values only
4. Pie chart

Answer 1 · Bar chart with summary statistic

34. Why can summary bars sometimes be insufficient for bivariate comparison?

1. They may hide variability and distribution within groups
2. They cannot show categories
3. They always exaggerate differences
4. They remove the need for sample sizes

Answer 1 · They may hide variability and distribution within groups

35. What does a negative association mean in a scatter plot?

1. As one variable increases, the other tends to decrease
2. Both variables increase together
3. The variables are identical
4. There are no outliers

Answer 1 · As one variable increases, the other tends to decrease

36. Why might jitter be added to a categorical-versus-numeric plot?

1. To reduce overlap and reveal point density
2. To change the category labels
3. To normalize the data
4. To remove missing values

Answer 1 · To reduce overlap and reveal point density

37. Which plot is often used to compare distributions of a numeric variable across categories?

1. Box plot
2. Single histogram only
3. Line chart without grouping
4. Map projection

Answer 1 · Box plot

38. What is a confounding concern in bivariate analysis?

1. An observed relationship may be influenced by a third variable
2. Two variables cannot be plotted together
3. Only categorical variables can confound
4. Confounding is visible only in pie charts

Answer 1 · An observed relationship may be influenced by a third variable

39. Why is sample size important when interpreting bivariate patterns?

1. Small samples can make patterns unstable or misleading
2. Large samples remove all bias
3. Sample size only matters in univariate analysis
4. It changes variable types

Answer 1 · Small samples can make patterns unstable or misleading

40. Which approach gives a fuller view than plotting only group means?

1. Showing distributions or individual data points as well
2. Removing all uncertainty information
3. Using only a table title
4. Sorting categories alphabetically every time

Answer 1 · Showing distributions or individual data points as well

41. What is one common way to encode an additional numeric variable in a scatter plot?

1. Marker size
2. Chart title length
3. Font family
4. Axis label rotation

Answer 1 · Marker size

42. Why can faceting be preferable to putting every variable into one plot?

1. It separates comparisons into manageable panels while preserving structure
2. It guarantees no information loss
3. It removes the need for scales
4. It works only for time series

Answer 1 · It separates comparisons into manageable panels while preserving structure

43. What does a heatmap often help show in multivariate analysis?

1. Patterns across two dimensions using color intensity
2. Only one variable distribution
3. Exact causal links
4. Database schema relationships

Answer 1 · Patterns across two dimensions using color intensity

44. What is a major tradeoff when adding more encodings to one chart?

1. More information can reduce readability
2. The chart becomes statistically invalid
3. The data type changes automatically
4. The need for labels disappears

Answer 1 · More information can reduce readability

45. Which method is useful for comparing the same relationship across subgroups?

1. Faceted scatter plots
2. Single pie chart
3. One histogram with no grouping
4. A decorative dashboard background

Answer 1 · Faceted scatter plots

46. Why should marker size be used carefully for quantitative encoding?

1. Humans judge area less precisely than position
2. Marker size cannot represent numbers
3. Large markers always improve clarity
4. It removes overplotting completely

Answer 1 · Humans judge area less precisely than position

47. What is the purpose of using hue for categories in a multivariate plot?

1. To distinguish groups visually
2. To sort data automatically
3. To remove legends
4. To convert categories into numbers

Answer 1 · To distinguish groups visually

48. Which issue can arise when too many categories are encoded by color?

1. The plot becomes hard to distinguish and interpret
2. The chart automatically becomes interactive
3. The axes disappear
4. The data becomes ordinal

Answer 1 · The plot becomes hard to distinguish and interpret

49. What can a pairwise plot matrix help with?

1. Exploring relationships among several numeric variables
2. Replacing all model evaluation
3. Showing only one variable at a time
4. Avoiding any preprocessing

Answer 1 · Exploring relationships among several numeric variables

50. Why is careful legend design important in multivariate charts?

1. Because multiple encodings can otherwise confuse interpretation
2. Legends always improve every chart equally
3. Legends replace the need for scales
4. Legends are only for maps

Answer 1 · Because multiple encodings can otherwise confuse interpretation

51. What should be the first priority in an explanatory chart?

1. A clear message
2. Maximum decoration
3. Showing every variable collected
4. Using the most advanced library

Answer 1 · A clear message

52. Why might an analyst choose a simpler chart for explanation even after using complex exploratory plots?

1. Simple charts often communicate the key point more effectively
2. Complex charts are never valid
3. Simple charts require no labels
4. Exploration and explanation must use identical visuals

Answer 1 · Simple charts often communicate the key point more effectively

53. Which title is better for an explanatory chart?

1. Customer retention fell sharply after month 6
2. Customer retention chart
3. Results
4. Data overview

Answer 1 · Customer retention fell sharply after month 6

54. What is the value of highlighting one series while muting others in an explanatory figure?

1. It directs attention to the intended comparison or takeaway
2. It removes the need for a legend in all cases
3. It guarantees objectivity
4. It increases the sample size

Answer 1 · It directs attention to the intended comparison or takeaway

55. Why are captions useful in explanatory communication?

1. They provide context, caveats, or interpretation
2. They replace the chart entirely
3. They are only decorative
4. They eliminate the need for titles

Answer 1 · They provide context, caveats, or interpretation

56. What is a danger of cherry-picking only supportive visuals?

1. It can misrepresent the full evidence and mislead the audience
2. It improves statistical validity
3. It removes bias from the analysis
4. It is required in executive reporting

Answer 1 · It can misrepresent the full evidence and mislead the audience

57. Which action best supports trustworthy storytelling with data?

1. Showing relevant context and limitations
2. Truncating axes without notice
3. Hiding contradictory evidence
4. Using unexplained acronyms everywhere

Answer 1 · Showing relevant context and limitations

58. Why might a slope chart be effective in some explanatory contexts?

1. It clearly emphasizes change between two points for multiple categories
2. It always replaces scatter plots
3. It is best for showing distributions
4. It removes the need for labels

Answer 1 · It clearly emphasizes change between two points for multiple categories

59. What is a good reason to annotate an outlier in a final chart?

1. To explain why it matters to the story or decision
2. To make the chart look busier
3. To avoid discussing the main trend
4. To replace the axis scale

Answer 1 · To explain why it matters to the story or decision

60. What best characterizes strong explanatory visualization?

1. Accurate, audience-aware, focused, and honest communication
2. Maximum complexity with minimal context
3. Heavy decoration with limited evidence
4. One chart type used for every message

Answer 1 · Accurate, audience-aware, focused, and honest communication